

Predicting Glycosylated Haemoglobin (Hba1c) and Determining Likelihood of Diabetes Diagnosis

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DATA309 Data Science Capstone Project

Project Brief

2.5% of Final Grade

Background and Domain

Diabetes is a chronic condition characterized by elevated blood sugar levels. It affects all ages and occurs in several forms including Type 1 Diabetes, Type 2 Diabetes, Prediabetes and Gestational Diabetes. Among these, Type 2 Diabetes is the most common, with 11% of the world population diagnosed. Symptoms include fatigue, blurred vision, unexplained weight loss and frequent urination. If left untreated, diabetes can cause many complications such as heart attack, stroke, skin infections, nerve damage and more (Cleveland Clinic, 2023). Due to its chronic nature, there is no cure to diabetes. Instead, treatments are focused on managing blood sugar levels and preventing complications by taking medicines such as insulin or developing healthy lifestyle habits. This project revolves around public healthcare and medical data science domain, which applies analytical and machine learning techniques to health-related data to support early diagnosis, treatment and health policy decisions.

Goal

The goal of this project is to improve early detection and monitoring of diabetes by building a predictive model using the data provided. The project aims to both predict the level of glucose level (HbA1c) and predict the likelihood of diabetes. We are to include the difference between male and female patients as well as how other variable affect these results. A gender-based analysis will be conducted to identify disparities in prediction accuracy and/or variable effects. If time is limited, the primary goal is to create a prediction of diabetes likelihood first since this has a clear decision-making impact.

Data

The data that we will use for this project is provided by the University of Canterbury via a GitHub repository. The dataset consists of 100,000 observations with 9 columns consisting numerical and categorical variables. Categorical variables include; gender, heart disease, hypertension, smoking history and diabetes status. Numerical variables include age, body mass index, glycosylated haemoglobin (HbA1c) levels and blood glucose levels.

Gender: Whether the participant is male or female

Heart disease: A condition that is closely related to having diabetes. 1 = Has heart disease, 0 = Doesn't have heart disease.

Hypertension (high blood pressure): A condition causing high pressure in the blood vessels, also closely related to having diabetes. 1 = Has hypertension, 0 = Doesn't have hypertension

Body mass index (BMI): Body fat percentage measured by weight and height. The higher the score, the higher the risk for diabetes.

Smoking history: Contains 5 categories; never, no info, current, former, ever, not current. Smoking is considered a risk factor for diabetes.

Diabetes status: Shows if the person has diabetes. 1 = Has diabetes, 0 = Doesn't have diabetes.

Age: The age of the participant

HbA1c levels: Measure of a person's blood sugar levels over the past 2-3 months.

Blood glucose levels: Amount of glucose in the blood stream at a given time.

Because we did not collect the data ourselves, we are unsure of the methods used to collect these. Therefore, challenges may arise in processing and analysis of the data. This can include biased samples such as imbalances in demographics which may skew the results. Poor quality of data, such as inconsistencies in data types, extremes and missing values can also lead to errors in the model fitting, ultimately causing incorrect diagnoses.

Planning

There are four major steps in completing this project: data collection, pre-processing, analysis, and evaluation. The time allotted for this project is 14 weeks including the two-week semester break. Within this timeframe, we have proposed a timeline based around the deadlines of the deliverables required for this project, namely the Project Brief, Project Report, and the Project Presentation.

Week 1 and 2, July 14 to July 27

- Clarifying the scope of the project
- Establishing methods of communication and tools to be used
- Prepare a project brief

Deliverables:

Project Brief, Due Date 3rd August

Weeks 3 to 5, July 28 to August 17 (Data Pre-processing)

- Pre-processing of data ie. cleaning, imputations
- Analyse data summaries, and create visualisations
- Data explorations

Weeks 6 to 9, August 18 to September 14 (Data Analysis)

- developing linear and logistic ANN regression models
- developing traditional logistic regression models
- conducting gender-based analyses

Weeks 10 to 13, September 15 to October 12 (Report Writing)

- Interpret results from analyses
- Compare results against other models, and studies by other academics
- Prepare a draft report
- Prepare for practice presentation

Deliverables:

Draft Project Presentation, Due 12th October

Draft Project Report, Due 12th October, 9pm

Week 14, October 13 to October 19

- final changes to project report based on supervisor's comments
- final preparations for project presentation

Deliverables:

Final Project Presentation, Due 16th October

Final Project Report, Due 19th October, 9pm

Strict adherence to the proposed plan is key to completing this project on time.

However, we plan on completing the tasks for the first five weeks as soon as possible.

This is because Artificial Neural Networks, or ANN, is an unfamiliar content to the group, therefore, finishing the earlier tasks will provide us more time to familiarise ourselves with the new content.

As with any other team projects, it is important for each member to communicate their progress to the group, as well as any issues or delays that the group needs to know. To manage this project we will be using Trello, a free online dashboard that will allow us to create lists of tasks, and assign member(s) into them. Each member are able to update the status their tasks via commenting and adding descriptions, so that other members are informed of each other's progress.

Every Wednesday during the semester, have also been scheduled for weekly meetups with the group and the project supervisors. The first few weeks will be introductory workshops to collaboration tools such as Google Colab and Git, and to Artificial Neural Networks. Subsequent meetings will involve updates on our current progress, as well as feedback on how to progress further.

The link below directs you to the Trello Board for this project.

<https://trello.com/invite/b/68799587cd5033b9d9b7623b/ATTI65c0a39fe7fcdfe4dbcb54c1ebe7b85cD9134E88/data309-diabetes-prediction>

Resources

To successfully complete this diabetes prediction project with the data mentioned above, several key resources are required across analysis tools and reference materials.

The tools that are available for this project will include:

- Coding language: R, Python, and Julia. However, team members are more familiar with R and Python, so these two will be used during the implementation. During the analysis, tasks will be split among members, and the language used is based on the members' preferences.
- Software and platform such as Jupyter Notebook, Google Colab, Github, VS Code, and RStudio
- GPU Access (optional): Request for a server to run the program if neural network models require high-performance computation.

Resources for reference and guidance on medical research, ANN theories, and the Carpentries platform will be utilized.

Constraints

The diabetes prediction project uses patients' sensitive health information, so it is subject to several important constraints, such as data security, accuracy, and ethics.

- Data Security:
 - Data must remain within the organization and cannot be shared with unauthorized users.
 - No use of external APIs for processing the dataset due to privacy commitments.
 - Access restrictions: Only authorized personnel are allowed to handle and analyze the dataset.
- Accuracy Limitations: certain factors may affect the precision of the level of glycosylated hemoglobin predictions that are not captured in the dataset
 - Biological factors such as hemoglobinopathies, medications, kidney or liver disease, and pregnancy.
 - Individual differences and ethnic differences.
 - Data quality and completeness
- Ethical Considerations:
 - Algorithmic bias must be acknowledged and minimized.
 - Fair use of the model - the system should not unfairly favor any group or individual
 - Data misuse is prohibited; the dataset and predictions must not be used outside of authorized purposes.

References:

- 1) Cleveland Clinic. (2023, February 17). *Diabetes*. Cleveland Clinic.
<https://my.clevelandclinic.org/health/diseases/7104-diabetes>